Equating equations (6.48) and (6.49),

$$\frac{L}{\sin(i-r)} = \frac{t}{\cos(r)}$$

After rearranging,

$$L = t\left(\frac{\sin(i-r)}{\cos(r)}\right) \qquad (6.50)$$

The lateral displacement depends upon (i) the thickness of the slab, (ii) the angle of incidence and (iii) the refractive index of the slab which decides the angle of refraction. Thicker the slab, larger will be the lateral displacement. Greater the angle of incidence, larger will be the lateral displacement. Higher the refractive index, larger will be the lateral displacement.

EXAMPLE 6.11

The thickness of a glass slab is 0.25 m. It has a refractive index of 1.5. A ray of light is incident on the surface of the slab at an angle of 60°. Find the lateral displacement of the light when it emerges from the other side of the glass slab.

Solution

Given, thickness of the slab, t = 0.25 m, refractive index, n = 1.5, angle of incidence, i = 60°.

Using Snell's law, 1 sin i = n sin r

$$\sin r = \frac{\sin i}{n} = \frac{\sin 60^\circ}{1.5} = 0.58$$

$$r = \sin^{-1}(0.58) = 35.25^\circ = 35^\circ 15' 0''$$

Lateral displacement is, $L = t\left(\frac{\sin(i-r)}{\cos(r)}\right)$

$$L = (0.25)\times\left(\frac{\sin(60-35.25)}{\cos(35.25)}\right) = 0.1281\,\text{m}$$

The lateral displacement is, $L = 12.81\,\text{cm}$

6.5 REFRACTION AT SINGLE SPHERICAL SURFACE

We have so far studied only the refraction at plane surface. The refraction can also take place at spherical surface between two transparent media. The laws of refraction hold good at every point on the spherical surface. The normal at the point of incidence is perpendicular drawn to the tangent plane of the spherical surface at that point. Therefore, the normal always passes through its center of curvature. The study of refraction at single spherical surface paves way to the understanding of thin lenses which consist of two refracting surfaces.

The following assumptions are made while considering refraction at spherical surfaces.

(i) The incident light is assumed to be monochromatic (single colour)

(ii) The incident light is very close to the principal axis (paraxial rays).

The sign conventions are similar to that of the spherical mirrors.

6.5.1 Equation for refraction at single spherical surface

Let us consider two transparent media with refractive indices n_1 and n_2 which are separated by a spherical surface as shown in Figure 6.31. Let C be the centre of curvature of the spherical surface. Let a point object O be in the medium n_1. The line OC is the principal axis that cuts the spherical surface at the pole P. As the rays considered are paraxial rays, the perpendicular dropped from the point of incidence to the principal axis is very close to the pole (or) passes through the pole itself.

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